Multi-Scale Vision Transformers for Dermoscopic Lesion Classification

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Abstract

Automated skin cancer screening requires discriminating subtle morphologic characteristics in cutaneous lesions. We propose a hierarchical multi-scale vision transformer architecture (MS-ViT) designed to capture fine-grained dermoscopic pigment network boundaries alongside global lesion symmetry. Evaluated on the ISIC-2020 archive comprising 33,126 dermoscopic images, MS-ViT achieved a top-1 balanced accuracy of 89.4% and an area under the ROC curve (AUC) of 0.942, outperforming the standard ResNet-50 baseline by 4.2%.

1. Introduction

Early diagnosis of malignant melanoma significantly improves 5-year patient survival rates. Recent advancements in deep convolutional neural networks (CNNs) have shown dermatologist-level classification capabilities in controlled experimental settings. However, conventional CNNs rely on localized receptive fields that often miss long-range architectural relationships across irregular lesion margins. Vision transformers provide self-attention mechanisms capable of modeling arbitrary spatial dependencies.

2. Methodology and Experimental Setup

The proposed MS-ViT pipeline partitions input images into multi-resolution patch tokens (16x16 and 8x8) to simultaneously encode microscopic cellular texture and macroscopic border irregularity.

Training Configuration:

- Optimizer: AdamW with weight decay 0.05
- Learning rate: 1e-4 with cosine learning rate decay
- Training duration: 100 epochs with batch size 64
- Resolution: All images standardized to 384x384 pixels
- Baselines: ResNet-50 and EfficientNet-B4 pretrained on ImageNet-1k

Dataset and Preprocessing:

We utilized the International Skin Imaging Collaboration (ISIC) 2020 dataset containing 33,126 dermoscopic images spanning 7 diagnostic classes: melanoma, melanocytic nevus, basal cell carcinoma, actinic keratosis, benign keratosis, dermatofibroma, and vascular lesion. Data augmentation included color jittering, random rotation (+/- 45 degrees), and subtle Gaussian blur to simulate real-world photographic variation.

3. Results and Empirical Findings
Evaluation was performed on an independent test partition of 4,500 held-out images.

Reported Results:
On the held-out ISIC validation set, our multi-scale transformer consistently improved classification accuracy across different lesion subtypes, achieving 91.2% sensitivity for melanoma detection and 89.4% overall balanced accuracy. The model demonstrated superior feature discriminability compared to ResNet-50 (85.2%) and EfficientNet-B4 (86.7%).

Subgroup Performance Disparity:
When disaggregating results across patient demographic groups, classification accuracy degraded by 6.8% when evaluated on dark skin phenotypes (Fitzpatrick skin types IV-VI). In these underrepresented cohorts, training representation accounted for less than 4% of total dataset samples, resulting in elevated false positive rates for benign pigmented lesions.

4. Limitations and Discussion
While the empirical results substantiate high discriminative capacity on standardized benchmarks, several critical limitations must be highlighted:

1. Acquisition Modality Dependency: The model was trained exclusively on epiluminescence contact dermoscopy. When evaluated on clinical macroscopic photography captured via consumer mobile smartphones, accuracy dropped sharply by 14.3% due to uncontrolled illumination, specular glare, and lack of specialized polarized lighting.

2. Computational Overhead: The multi-scale self-attention mechanism requires 22.4 GFLOPs per forward pass, resulting in 4.3x higher latency compared to MobileNetV3. This latency presents deployment challenges for low-power handheld clinical devices in rural clinics.

3. Lack of Prospective Validation: The retrospective design does not account for clinician interaction dynamics, biopsy thresholds, or multi-lesion clinical prioritization.